

APPENDIX A - CORE PRACTICE

TIERED SCAFFOLDING

SOLUTIONS IN APPENDIX B

{{ CHAPTER_TITLE: Comprehensive Topic Mastery & Practice Items }}

TIER 1: GUIDED PRACTICE

ITEM: {{ ITEM_ID: PRAC-01 }}

{{ CAPABILITY_REF: CAP-ASSIGN-OXIDATION-NUMBER-CORE }}

{{ PROMPT }}: Determine the oxidation state of Phosphorus in H_3PO_4 .
Representation Spec: FORMULA_ANATOMY_VIEW

Step-by-Step Guided Scaffolding:

1. Identify fixed-rule elements: Hydrogen is in non-metal compound $\Rightarrow \text{H} = +1$.
Three Hydrogen atoms contribute: $3 \times (+1) = +3$.
2. Identify Oxygen default: No peroxide bond present $\Rightarrow \text{O} = -2$.
Four Oxygen atoms contribute: $4 \times (-2) = -8$.
3. Set up the Sum Invariant (Net charge = 0):
 $(+3) + \text{P} + (-8) = 0 \Rightarrow \text{P} - 5 = 0 \Rightarrow \text{P} = [\quad]$.

Solution Reference: [See Appendix B --> SOL-PRAC-01]

TIER 2: FADED PRACTICE

ITEM: {{ ITEM_ID: PRAC-02 }}

{{ CAPABILITY_REF: CAP-TRACK-ELEMENT-CHARGE-CHANGE }}

{{ PROMPT }}: For the reaction: $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$,
identify which element undergoes oxidation and which undergoes reduction.

Faded Support Steps:

1. Write ON above each atom in reactants: $\text{Zn} = [\quad]$, $\text{H in HCl} = [\quad]$, $\text{Cl in HCl} = [\quad]$.
2. Write ON above each atom in products: $\text{Zn in ZnCl}_2 = [\quad]$, $\text{Cl} = [\quad]$, $\text{H in H}_2 = [\quad]$.
3. Track direction of change:
* Zn changes from $\underline{\quad}$ to $\underline{\quad} \Rightarrow$ (Increase/Decrease) $\Rightarrow [\quad]$.
* H changes from $\underline{\quad}$ to $\underline{\quad} \Rightarrow$ (Increase/Decrease) $\Rightarrow [\quad]$.

Solution Reference: [See Appendix B --> SOL-PRAC-02]

TIER 3: INDEPENDENT PRACTICE

ITEM: {{ ITEM_ID: PRAC-03 }}

{{ CAPABILITY_REF: CAP-ATTACH-REDOX-ROLES }}

{{ PROMPT }}: In the industrial reaction: $\text{Fe}_2\text{O}_3\text{(s)} + 3\text{CO(g)} \rightarrow 2\text{Fe(s)} + 3\text{CO}_2\text{(g)}$:
(a) Name the oxidising agent and reducing agent.
(b) Calculate the total moles of electrons transferred per mole of Fe_2O_3 reacted.

Instructions: Complete without hints. Show your work clearly.

Your Working Area:

* Oxidising Agent: _____ Reducing Agent: _____

* Electron Transfer Calculation: _____

Solution Reference: [See Appendix B --> SOL-PRAC-03]

TIER 4: DIAGNOSTIC PROBE (TRAP DETECTOR)

ITEM: {{ ITEM_ID: PRAC-04 }}

{{ CAPABILITY_REF: CAP-DISCRIMINATE-PEROXIDE-EXCEPTION }}

{{ PROMPT }}: A student calculates the oxidation state of Oxygen in BaO_2 by applying the standard rule ($\text{O} = -2$), concluding $\text{Ba} = +4$. Evaluate this claim.

Diagnostic Evaluation Questions:

1. Is $\text{Ba} = +4$ physically permissible for an Alkaline Earth metal (Group 2)? [YES / NO]
2. Which rule takes priority: Group 2 fixed charge (+2) OR default Oxygen (-2)?
Priority Rule: _____
3. Deduce the correct oxidation state of Oxygen in barium peroxide (BaO_2):
Correct ON of Oxygen = [$\underline{\quad}$].

Solution Reference: [See Appendix B --> SOL-PRAC-04]